

Data Driven Channel Characterization of Human Body Communication

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ABSTRACT

Intra-body communication (IBC) is a novel research area that will foster personalized medicine by allowing the interconnection of implanted devices through energy-efficient technologies, such as the galvanic coupling (GC) technique considered in this work. Channel characterization of the human body is here evaluated through experimental data considering in-vivo tissues in the frequency range up to 100 kHz. Pseudorandom noise (PN) sequences are transmitted in baseband and a correlative channel sounding system is implemented to evaluate the channel impulse and frequency response through real measurements.

CCS CONCEPTS

• **Hardware** → *Digital signal processing; Sound-based input / output; Sensor devices and platforms*; • **Wireless devices**; • **Networks** → *Physical links*.

KEYWORDS

Intra-body networks, galvanic coupling, coupling circuits, wireless sensor networks, biomedical engineering, experimental testbed

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1 INTRODUCTION

Next-generation healthcare with real-time monitoring of physiological data will be assured by novel energy-efficient communication techniques able to interconnect implanted devices, such as the galvanic coupling (GC) technology [1]. GC employs a pair of electrodes as transmitter and receiver, which may be attached or implanted in the body, conveying weak (<1 mA) electrical currents modulated with data [2] at low/medium frequencies (1 KHz-100MHz).

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Previous papers on GC analyzed the body channel model mainly using finite-element methods and equivalent circuit-analysis-based modeling [1]. A limited amount of research was spent on defining a channel impulse response (CIR) to characterize the body channel [3], which is instead fundamental to the design of a communication system. In this work we use a correlative channel sounding method [3, 4] employing pseudorandom noise (PN) sequences to evaluate the GC channel impulse response through experimental measurements. While [3] considered GC frequencies in the range 100 KHz - 1 MHz, we evaluate the band up to 100 KHz, so that we can transmit the signals in baseband, avoiding any modulation scheme and complex synchronization method at the receiver, which is suitable for implanted devices. The goal is to characterize the GC body channel at these frequencies, so that the experimental findings will be useful to develop future ultra wide band (UWB) transmission techniques operating at very low power transmission with simple implanted transceivers. Moreover, we specifically evaluate the CIR considering in-vivo tissues.

2 CHANNEL MODELING

In this work a correlative channel sounder is leveraged to analyze the body channel behavior. As is well known, the received signal may be described as $y(t) = x(t) * h(t) + n(t)$, where $x(t)$ and $y(t)$ are the transmitted and received signal, $h(t)$ is the impulse response of the channel, $n(t)$ is the additive noise and $*$ is the convolution operator. Correlating each side of the above equation with $x(t)$, we obtain:

$$R_{xy}(\tau) = h(\tau) * R_{xx}(\tau) \quad (1)$$

where R_{xy} is the cross-correlation function of $x(t)$ and $y(t)$, $R_{xx}(\tau)$ is the auto-correlation function of $x(t)$, τ is the delay time, and $n(t)$ and $x(t)$ are assumed uncorrelated. If the channel impulse response $h(t)$ varies slowly within the time interval needed to measure the correlation function, then (1) can be employed to measure $h(t)$, as long as R_{xx} approaches a delta function. This can be achieved using PN sequences as transmitted signal $x(t)$ and correlating the received signal with the transmitted PN sequence in (1) to calculate the cross-correlation R_{xy} [4], which corresponds to the CIR. The capability to change bandwidth and maximum delay are important features of PN sequence sounders. Moreover, the sounder exhibits a processing gain, which reduces transmitter power requirements, and may be increased by employing longer code words. Using maximal-length PN sequences as transmitted signal leads to an auto-correlation function featuring a high correlation peak and very low side lobes. This characteristic enables the detection of each multi-path component at the receiver when correlating the

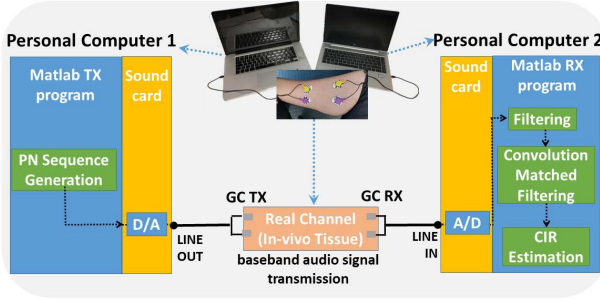


Figure 1: Implemented GC Testbed

channel output with the originally transmitted PN sequence by means of a convolution matched filter [3].

3 EXPERIMENTAL METHOD AND CHANNEL CHARACTERIZATION

To measure the channel impulse responses, PN sequences were transmitted in baseband. A linear generated polynomial PN sequence of degree $m = 14$ was implemented using a linear-feedback shift register with a chip duration of $5.2 \mu\text{s}$ (≈ 100 kHz bandwidth). The testbed [5] was modified to implement a correlative channel sounding (Fig. 1). The system requires limited hardware since it uses only two PCs with sound cards to support signal transmission/reception in a subset of the GC frequency range [5]. We use battery-powered PCs, unplugged from the grid to avoid common ground return paths between transmitter and receiver, as required by the GC. The generated PN sequences are transmitted using a Matlab program, and then converted from digital to analog (D/A) domain to be sent over the sound card of the transmitter. The signal leaves the PC to be injected into the biological tissue by means of a cable connected to the *LINE OUT* jack on one side and to two electrodes the GC transmitter on the opposite side. After transmission, the two receiving electrodes detect the received signal, which is sent to the other PC through a cable connected to the *LINE IN* jack. Data processing at the receiver includes a 50 Hz filter and a convolution matched filter to correlate the channel output with the transmitted PN sequence known at the receiver, to build the CIR estimation. The audio frequency sampling f_{sa} is set to 192 KHz with 16 bits per sample. The experiments were carried out on a human leg, consisting of skin, fat, muscle and bone tissues (Fig. 1).

The measured CIR is reported in Fig. 2 (a) and (c) for different values of delay τ , normalized to the sampling interval $1/f_{sa}$, and of electrode sizes (0.5 mm and 1 cm). The figures reveal the high peak-to-off peak ratio discussed in Sec. II, which leads to good correlation results from the experiments. All CIRs obtained from the experiments for different distances show a similar impulse response, indicating absence of multiple paths in the channel environment. The corresponding frequency domain representation, the channel frequency response (CFR), for an assumed transmitter bandwidth of 96 kHz and for frequency normalized to f_{sa} , indicates that the channel is relatively flat within the frequency range of interest with higher attenuation for smaller electrode sizes (Fig. 2 (b) and (d)).

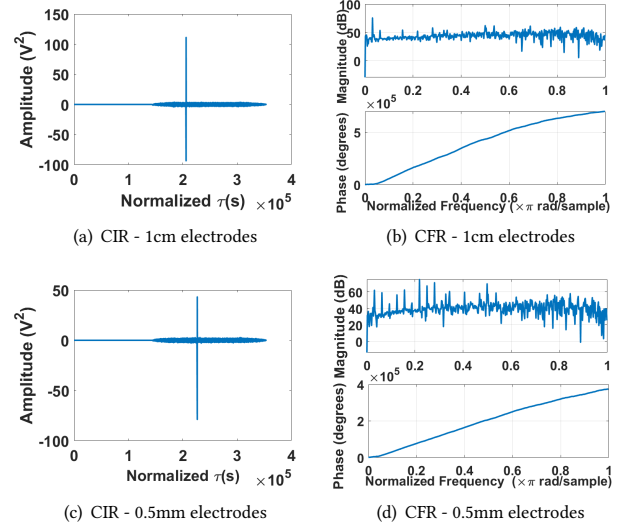


Figure 2: The measured channel impulse response (CIR) and channel frequency response (CFR) for wearable configuration with in-vivo tissues (distance $d_{tx-rx} = 6$ cm)

4 CONCLUSION

This paper describes the experiments conducted to characterize the GC body channel. Results indicate that the frequency channel response is relatively flat across the range of interest. Future work includes (i) analysis and comparison of experimental results with analytical models considering different locations of electrodes, and (ii) development of UWB and compressive sensing transmission methods [6, 7], suitable for simple implanted transceivers. Also, multiple implants scenario are envisioned, in which it is essential to develop opportunistic wake-up methods with location awareness of devices [8–10] or to exploit resource allocation methods [11] to minimize energy consumption as required in intra-body networks.

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